

Q1

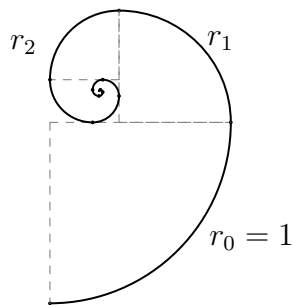
Let

$$\phi = \frac{1 + \sqrt{5}}{2}.$$

An approximation to the *golden spiral* is constructed from infinitely many consecutive quarter-circle arcs. The n th arc has radius r_n , where

$$r_0 = 1 \quad \text{and} \quad r_n = \phi r_{n+1} \quad \text{for } n = 0, 1, 2, \dots$$

The arcs are joined end-to-end as shown. The diagram is not drawn to scale.



What is the total length of the spiral?

- A $\frac{\pi}{2\phi}$
- B $\frac{\pi}{2}$
- C $\frac{\pi\phi}{2}$
- D $\frac{\pi\phi^2}{2}$
- E The length is infinite.

Q2

Let $f: [a, b] \rightarrow \mathbb{R}$, where $a < b$. Given that the statement

If a real-valued function f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists some $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

is true, which of the following statements must be true?

- A** If there is no $c \in (a, b)$ satisfying the displayed equation, then f is not continuous on $[a, b]$ or f is not differentiable on (a, b) .
- B** If there is no $c \in (a, b)$ satisfying the displayed equation, then f is not continuous on $[a, b]$ and f is not differentiable on (a, b) .
- C** If f is not continuous on $[a, b]$ or is not differentiable on (a, b) , then there is no $c \in (a, b)$ satisfying the displayed equation.
- D** If there exists some $c \in (a, b)$ satisfying the displayed equation, then f is continuous on $[a, b]$ and differentiable on (a, b) .
- E** f is continuous on $[a, b]$ and differentiable on (a, b) , but there is no $c \in (a, b)$ satisfying the displayed equation.

Q3

Exactly one of the following statements is true. Which one?

I $x^2 + y^2 \geq 1$

II $x^2 + y^2 < 1$

III $\left(x^2 - \frac{1}{2}\right)\left(y^2 - \frac{1}{2}\right) > 0$ and $|x| \leq \frac{\sqrt{2}}{2}$

IV $(x, y) = (1, 1)$

V $|x| + |y| \leq 2$

A I

B II

C III

D IV

E V

Q4

A polynomial is given by

$$P(x) = x^4 + ax^2 + b$$

Which of the following is a necessary but not sufficient condition for $P(x)$ to have two distinct real roots?

I $b \leq 0$

II $a^2 > 4b$

III $a^2 \geq 4b$

IV $a \leq 0$

A None of them

B I only

C II only

D III only

E I and II

F I and III

G II and III

H I, II and III

I I, II and IV

J I, III and IV

K I, II, III and IV

Q5

Evaluate the integral

$$\int_{-10}^{10} |2x - \lfloor x \rfloor| \, dx$$

Where $\lfloor x \rfloor$ is equal to the greatest integer less than or equal to x .

- A** 0
- B** 10
- C** 45.5
- D** 50
- E** 55
- F** 95.5
- G** 100
- H** 100.5

Q6

Consider the following condition on a set S :

A positive integer x is an element of S only if for every $y \in S$ for which at least one of $x + y$ and $|x - y|$ is prime, both $|x - y|$ and $x + y$ are prime.

Which of the following sets does not satisfy the condition?

- A $\{2, 5, 9, 15, 39\}$
- B $\{2, 5, 9, 15, 21\}$
- C $\{4, 7, 15, 27, 33\}$
- D $\{6, 11, 17, 23, 43\}$
- E $\{6, 11, 17, 23, 37\}$
- F $\{5, 8, 15, 21, 39\}$

Q7

Let

$$f(x) = x + \frac{4}{x}, \quad -2 < x < 2, \quad x \neq 0.$$

A student gives the following argument:

(I) Differentiating gives

$$f'(x) = 1 - \frac{4}{x^2}.$$

(II) For every x in the domain, $0 < x^2 < 4$, so $f'(x) < 0$.

(III) Consequently, whenever $a < b$ are in the domain, $f(a) > f(b)$.

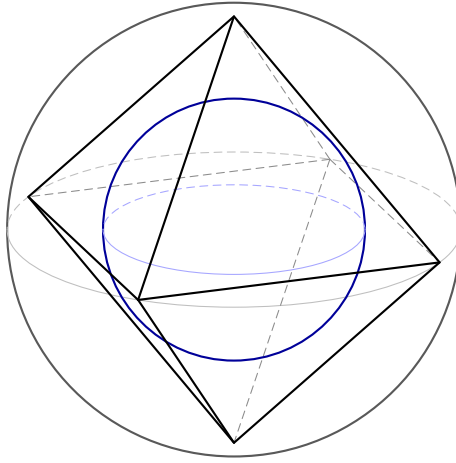
(IV) Therefore, the equation $f(x) = k$ has at most one solution in the domain for every real number k .

Which statement correctly assesses the argument?

- A Step (I) is incorrect because the derivative of $4/x$ is $4/x^2$.
- B Step (II) is incorrect because multiplying an inequality by x^2 may reverse its direction.
- C Step (III) is incorrect, and the conclusion in Step (IV) is false.
- D Step (III) is incorrect, but the conclusion in Step (IV) is true.
- E Steps (I), (II), (III) are correct, but Step (IV) does not follow from them.
- F The whole argument is correct.

Q8

A regular octahedron is inscribed inside a sphere of radius 1, with all six of its vertices touching the sphere. A sphere is then inscribed in the octahedron tangent to all 8 faces of the octahedron. This process is infinitely repeated. What is the sum of the surface areas of all spheres?



- A 2π
- B 4π
- C 6π
- D 8π
- E $(6 + 2\sqrt{3})\pi$
- F 10π

Q9

The expansion of $(x\sqrt[4]{17} + y\sqrt[6]{18})^k$ has n terms with rational coefficients.

Given that $10 \leq k \leq 20$ and k is a positive integer, which of the following statements are true?

- I** if k is even, then $n = 1$
- II** if k is a multiple of 4, then $n = 1$
- III** if k is odd, then $n = 0$

- A** I only
- B** II only
- C** III only
- D** I and II
- E** II and III
- F** I and III
- G** I, II and III
- H** None of the statements are true

Q10

Real numbers a and b satisfy the equations

$$a^b = 5^{5^{18}} \text{ and } (\log_5 a)^{\log_5 b} = 5^{17}$$

What is the sum of the possible values of a + the sum of the possible values of b ?

A 17

B 18

C 34

D 36

E $5^{5^{17}} + 5^5 + 5^{-1} + 5^{-17}$

F $5 + 5^5 + 5^{17} + 5^{5^{17}}$

G $10 + 2 \times 5^{17}$

H $2 \times 5^5 + 2 \times 5^{17}$

Q11

A fair six sided dice is tossed 17 times, with each toss independent of each other. Let P_1 be the probability that the dice land on each of 1, 2, 3, 4 and 5 exactly three times, and let P_2 be the probability that the dice land on 1 exactly once, 2 exactly twice, 3 exactly three times, 4 exactly four times, and 5 exactly five times. What is $P_1 : P_2$?

- A 40 : 9
- B 9 : 40
- C 3 : 1
- D 1 : 3
- E 7 : 2
- F 2 : 7
- G 16 : 9
- H 9 : 16

Q12

Consider the following statements about a function $f(x)$.

- I** If $f(x)$ is strictly increasing and differentiable in $\mathbb{R}$, then $f'(x) > 0$ for every real x .
- II** If $f(x) > g(x)$ for every real x , then $f'(x) > g'(x)$ for every real x , provided both functions are differentiable.
- III** If $f(x)$ is strictly increasing on $[0, 1]$, $f(0) = 0$ and $f(1) = 2$, then $f(c) = 1$ for some $c \in (0, 1)$.

Which of the statements are true?

- A** I only
- B** II only
- C** III only
- D** I and II
- E** II and III
- F** I and III
- G** I, II and III
- H** None of them

Q13

Consider the statement

If a sequence satisfies $a_{n+1} < a_n$ and $a_n > 0$ for all n , then $a_n \rightarrow 0$

Which of the following provides a counterexample to the statement?

A $a_{n+1} - a_n = \frac{1}{n(n+1)}, a_1 = -1$

B $a_{n+1} - a_n = \frac{1}{n(n+1)}, a_1 = 0$

C $a_{n+1} - a_n = -\frac{1}{n(n+1)}, a_1 = 1$

D $a_{n+1} - a_n = -\frac{1}{n(n+1)}, a_1 = 2$

E $a_{n+1} - a_n = -\frac{1}{n(n+1)}, a_1 = -1$

F There are no counterexamples and the statement is true

Q14

Consider the equation

$$\lfloor x - y \rfloor = \lfloor x^2 + y^2 \rfloor$$

Given that $x^2 + y^2 \leq 1$, what is the area of the region in the cartesian plane containing the set of points (x, y) satisfying both conditions?

A $\frac{\pi}{4} + \frac{1}{2}$

B $\frac{\pi}{4}$

C 1

D $\frac{\pi}{2} + \frac{1}{2}$

E $\frac{\pi}{2}$

F $\frac{\pi}{8} + 1$

G $\frac{\pi}{8} + \frac{1}{4}$

Q15

Five mathematicians, Archimedes, Bernoulli, Cantor, Descartes, and Euler, make the following statements:

Archimedes: Bernoulli is telling the truth

Bernoulli: If Cantor is telling the truth, then Descartes is also telling the truth

Cantor: If Euler is lying, then Descartes is also lying

Descartes: If Archimedes is lying, then Euler is telling the truth

Euler: Cantor is lying

Given that each mathematician either always tells the truth or is always lying, then which of the following is necessarily true?

- A Archimedes is telling the truth
- B Bernoulli is telling the truth
- C Exactly one of Cantor and Descartes is telling the truth
- D Descartes is telling the truth
- E Euler is telling the truth
- F If Cantor is telling the truth, then so is Descartes

Q16

For a real number k , consider the statements:

- I** For every $x \in [0, 1]$, there exists $y \in [0, 1]$ such that $(x - y)^2 \geq k$.
- II** There exists $y \in [0, 1]$ such that, for every $x \in [0, 1]$, $(x - y)^2 \geq k$.

For which values of k is statement I true and statement II false?

- A** $k < 0$
- B** $k = 0$
- C** $0 \leq k \leq \frac{1}{4}$
- D** $0 < k \leq \frac{1}{4}$
- E** $\frac{1}{4} < k \leq 1$
- F** There are no such values of k

Q17

A student wishes to prove that, for a positive integer n ,

$$n^4 - 1$$

is divisible by 240.

They consider the following possible hypotheses:

1. n is prime and $n > 5$.
2. n is not divisible by 2, 3 or 5.
3. n leaves remainder 1 or 239 when divided by 240.

Which statement is correct?

- A** Only I is sufficient.
- B** Only II is sufficient.
- C** I and II are sufficient, but III is not.
- D** All three are sufficient, but only II is necessary.
- E** All three are sufficient, but none is necessary.
- F** All three are necessary and sufficient.

Q18

A sequence is defined by

$$u_1 = a, \quad u_{n+1} = u_n^2 - 2,$$

where a is real.

A student claims:

If $a > 2$, then the sequence is strictly increasing, so it is unbounded.

Although the conclusion is correct, the reason given is insufficient.

Which condition on a is **necessary and sufficient** for the sequence to be bounded, meaning that there exists $M > 0$ such that $|u_n| \leq M$ for every positive integer n ?

- A $a \leq 2$
- B $-2 < a < 2$
- C $-2 \leq a \leq 2$
- D $-1 \leq a \leq 2$
- E $a \in \{-2, -1, 2\}$

Q19

For a real number a , consider the following statements:

- I. For every real number x , there exists a real number $y \in [1, 2]$ such that

$$x^2 - 2xy + ay^2 \geq 0.$$

- II. There exists a real number $y \in [1, 2]$ such that, for every real number x ,

$$x^2 - 2xy + ay^2 \geq 0.$$

For which values of a is statement I true and statement II false?

A $0 < a < 1$

B $\frac{3}{4} \leq a < 1$

C $\frac{8}{9} < a < 1$

D $\frac{8}{9} \leq a < 1$

E $\frac{8}{9} \leq a \leq 1$

F There are no such values of a .

Q20

Let

$$P(x) = x^5 + ax^3 + bx + c,$$

where a, b, c are real constants.

Consider the statement:

For all real numbers u, v , if $P(u)P(v) \leq 0$, then $uv \leq 0$.

Which condition is **necessary and sufficient** for this statement to be true?

A $c = 0$ and $b \geq 0$.

B $c = 0$, $a \geq 0$ and $b \geq 0$.

C $c = 0$ and $b > \frac{a^2}{4}$.

D $c = 0$, and either $a \geq 0$ and $b \geq 0$, or $a < 0$ and $b \geq \frac{a^2}{4}$.

E $c = 0$, and either $a \geq 0$ and $b \geq 0$, or $a < 0$ and $b > \frac{a^2}{4}$.

F $c = 0$, and either $a > 0$ and $b > 0$, or $a \leq 0$ and $b > \frac{a^2}{4}$.